

REQUIRE-X ENGINEERING REPORT

Project: REQUIRE-X Analysis Report: ecommerce_srs.txt | Timestamp: 2026-08-27 16:03:19 | Framework: Multi-Agent AI

1. Executive Summary

REQUIRE-X Multi-Agent AI Framework evaluated the Software Requirement Specification 'ecommerce_srs.txt'. The specification comprises 11 total requirements (5 Functional, 6 Non-Functional) with an estimated agile effort of 43 story points. The document attained an ISO/IEC/IEEE 29148:2018 compliance score of 92.5% (Grade: A). The framework generated 16 automated test cases, achieving 100.0% bidirectional traceability coverage mapped to a recommended Event-Driven Microservices Architecture blueprint.

2. Requirements Engineering Quality Metrics

Metric	Value	Metric	Value
Total Requirements	11	ISO 29148 Compliance	92.5% (A)
Functional Requirements (FR)	5	Ambiguity Rate	18.2% (2 flaws)
Non-Functional Requirements (NFR)	6	Traceability Coverage	100.0%
Total Dependencies	4	Estimated Story Points	43 pts

3. ISO/IEC/IEEE 29148:2018 Standards Compliance Audit

ISO 29148 Characteristic	Score	Status	Remediation / Finding
Completeness	90%	Pass	Ensure all edge cases, exceptions, and security guardrails are fully specified.
Consistency	92%	Pass	Maintain a centralized data dictionary to avoid conflicting entity definitions.
Unambiguity	86%	Pass	Apply suggested rewrites to eliminate subjective qualifiers and quantify all performance bounds.
Verifiability / Testability	100%	Pass	Define numerical acceptance metrics (e.g. latency in ms, uptime %).
Modifiability	95%	Pass	Keep requirements isolated to minimize cross-cutting modification side-effects.
Traceability	96%	Pass	Integrate IDs with Jira/ALM tools for end-to-end forward traceability.
Feasibility	90%	Pass	Conduct technical spike tests on high-complexity modules.
Correctness	91%	Pass	Review with stakeholders during formal SRS walkthrough.
Necessity	94%	Pass	Validate low-priority features against project budget.

4. Classified Requirements Catalog

ID	Type	Category	Statement	Pts
FR-001	Functional	Payment & Billing	The system shall enable registered consumers to browse product catalogs, filter by category/price, and perform full-text fuzzy searches.	3
FR-002	Functional	Payment & Billing	The system shall manage persistent digital shopping carts with automatic price synchronization and inventory reservation locks.	3
FR-003	Non-Functiona l	Compliance	The system shall process credit card, digital wallet, and UPI transactions via PCI-DSS certified payment gateways.	3
FR-004	Functional	Payment & Billing	The system shall calculate localized sales taxes, international customs duties, and courier shipping rates dynamically during checkout.	3
FR-005	Functional	Notifications	The system shall notify customers via automated email and SMS upon order placement, shipment dispatch, and delivery.	8
FR-006	Non-Functiona l	Performance	The system shall allow merchant sellers to upload inventory spreadsheets, update SKU pricing, and manage return requests.	3
FR-007	Functional	Core Business Logic	The application should be seamless and work efficiently across web and mobile browsers.	3
NFR-001	Non-Functiona l	Performance	The platform architecture shall auto-scale horizontally to support 25,000 requests per second during flash sale events without service disruption.	3
NFR-002	Non-Functiona l	Security	User passwords shall be hashed using Argon2id and API endpoints shall enforce OAuth 2.0 JWT bearer token authentication.	3
NFR-003	Non-Functiona l	Performance	The product catalog search endpoint shall return results in less than 150 milliseconds for 99% of requests.	3
NFR-004	Non-Functiona l	Reliability	The checkout and order placement subsystems shall guarantee 99.99% high availability with zero single points of failure.	8

5. Ambiguity Detection & Disambiguation Rewrites

Req ID	Defect Type	Flawed Phrase / Explanation	ISO Disambiguated Rewrite
FR-007	Subjective Language (Warning)	'seamless' 'Seamless' is a marketing buzzword. Detail exact API integration protocols and error mitigation.	The application should be using authenticated RESTful JSON APIs with automatic token refresh and work efficiently across web and mobile browsers.
NFR-002	Passive Voice / Missing Actor (Minor)	'be hashed' Passive construct 'be hashed' obscures the system actor responsible for the action.	The System shall explicitly execute the action rather than stating 'be hashed'.

6. Software Architecture Recommendation

Recommended Pattern: Event-Driven Microservices Architecture

Rationale: Selected Event-Driven Microservices architecture due to demanding non-functional requirements in Scalability, high throughput, and independent domain decoupling. Asynchronous messaging isolates failure domains and supports horizontal elastic autoscaling.

7. Requirements Traceability Matrix (RTM)

Req ID	Requirement Title	Architectural Module	Mapped Test Cases	Coverage
FR-001	Enable registered consumers to browse product	Core Business Logic Engine	TC-001, TC-002	Covered
FR-002	Manage persistent digital shopping carts with	Core Business Logic Engine	TC-003, TC-004	Covered
FR-003	Process credit card, digital wallet, and	Telemetry, Monitoring & Quality Guard	TC-005	Covered
FR-004	Calculate localized sales taxes, international customs	Core Business Logic Engine	TC-006, TC-007	Covered
FR-005	Notify customers via automated email and	Core Business Logic Engine	TC-008, TC-009	Covered
FR-006	Allow merchant sellers to upload inventory	Telemetry, Monitoring & Quality Guard	TC-010	Covered
FR-007	Be seamless and work efficiently across	Core Business Logic Engine	TC-011, TC-012	Covered
NFR-001	The platform architecture shall auto-scale horizontally	Telemetry, Monitoring & Quality Guard	TC-013	Covered
NFR-002	User passwords shall be hashed using	API Gateway & Security Provider	TC-014	Covered
NFR-003	The product catalog search endpoint shall	Telemetry, Monitoring & Quality Guard	TC-015	Covered
NFR-004	The checkout and order placement subsystems	Telemetry, Monitoring & Quality Guard	TC-016	Covered

8. Synthesized Test Suite (Sample)

TC-001 [Functional] — Verify Successful Execution of Enable registered consumers to browse product (Maps to: FR-001)

Preconditions: System is active, database is connected, user has valid role permissions.

Steps: 1. Navigate to the interface corresponding to 'Enable registered consumers to browse product'. 2. Supply valid operational input parameters. 3. Trigger action defined in 'The system shall enable registered consumers to browse product catalogs, filter by category/price, and perform full-text fuzzy searches.'. 4. Inspect system response and database state.

Expected Result: System successfully executes Enable registered consumers to browse product with HTTP 200/201 status and persists expected state.

TC-002 [Boundary / Edge Case] — Negative Validation & Boundary Check for Enable registered consumers to browse product (Maps to: FR-001)

Preconditions: System is active.

Steps: 1. Trigger 'Enable registered consumers to browse product' with malformed, empty, or out-of-range input data. 2. Attempt action with expired authentication token or unauthorized role. 3. Inspect error telemetry and user feedback.

Expected Result: System safely rejects malformed input with user-friendly error message (HTTP 400/403/422) without crashing.

TC-003 [Functional] — Verify Successful Execution of Manage persistent digital shopping carts with (Maps to: FR-002)

Preconditions: System is active, database is connected, user has valid role permissions.

Steps: 1. Navigate to the interface corresponding to 'Manage persistent digital shopping carts with'. 2. Supply valid operational input parameters. 3. Trigger action defined in 'The system shall manage persistent digital shopping carts with automatic price synchronization and inventory reservation locks.'. 4. Inspect system response and database state.

Expected Result: System successfully executes Manage persistent digital shopping carts with with HTTP 200/201 status and persists expected state.

TC-004 [Boundary / Edge Case] — Negative Validation & Boundary Check for Manage persistent digital shopping carts with (Maps to: FR-002)

Preconditions: System is active.

Steps: 1. Trigger 'Manage persistent digital shopping carts with' with malformed, empty, or out-of-range input data. 2. Attempt action

with expired authentication token or unauthorized role. 3. Inspect error telemetry and user feedback.

Expected Result: System safely rejects malformed input with user-friendly error message (HTTP 400/403/422) without crashing.